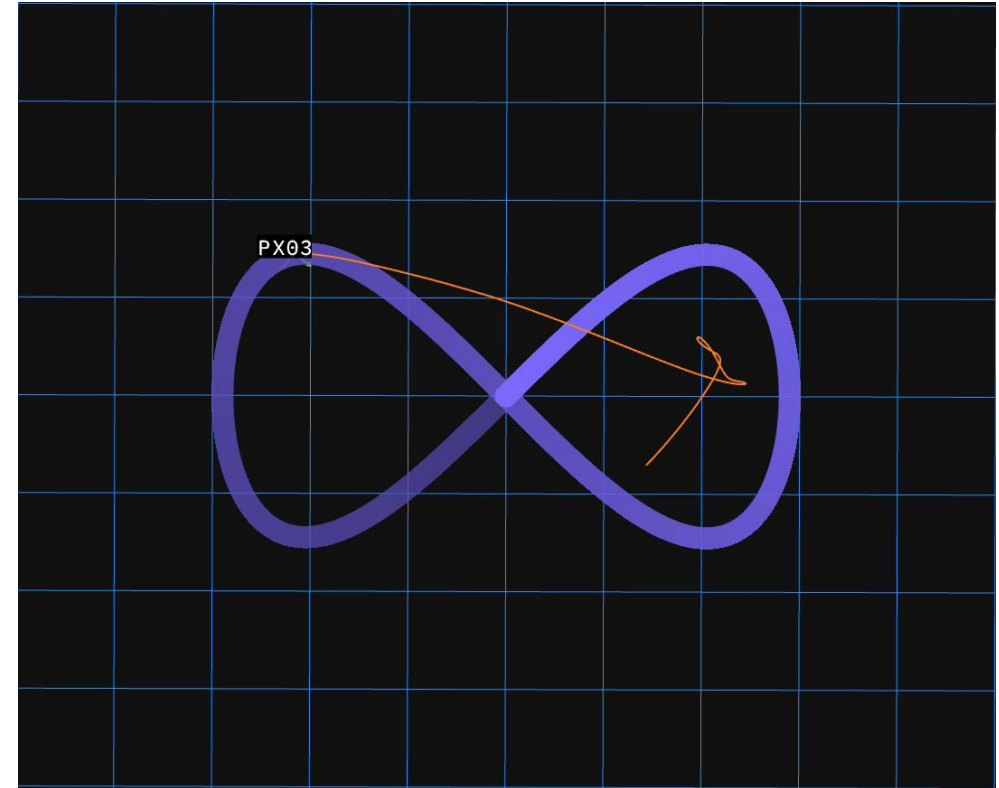
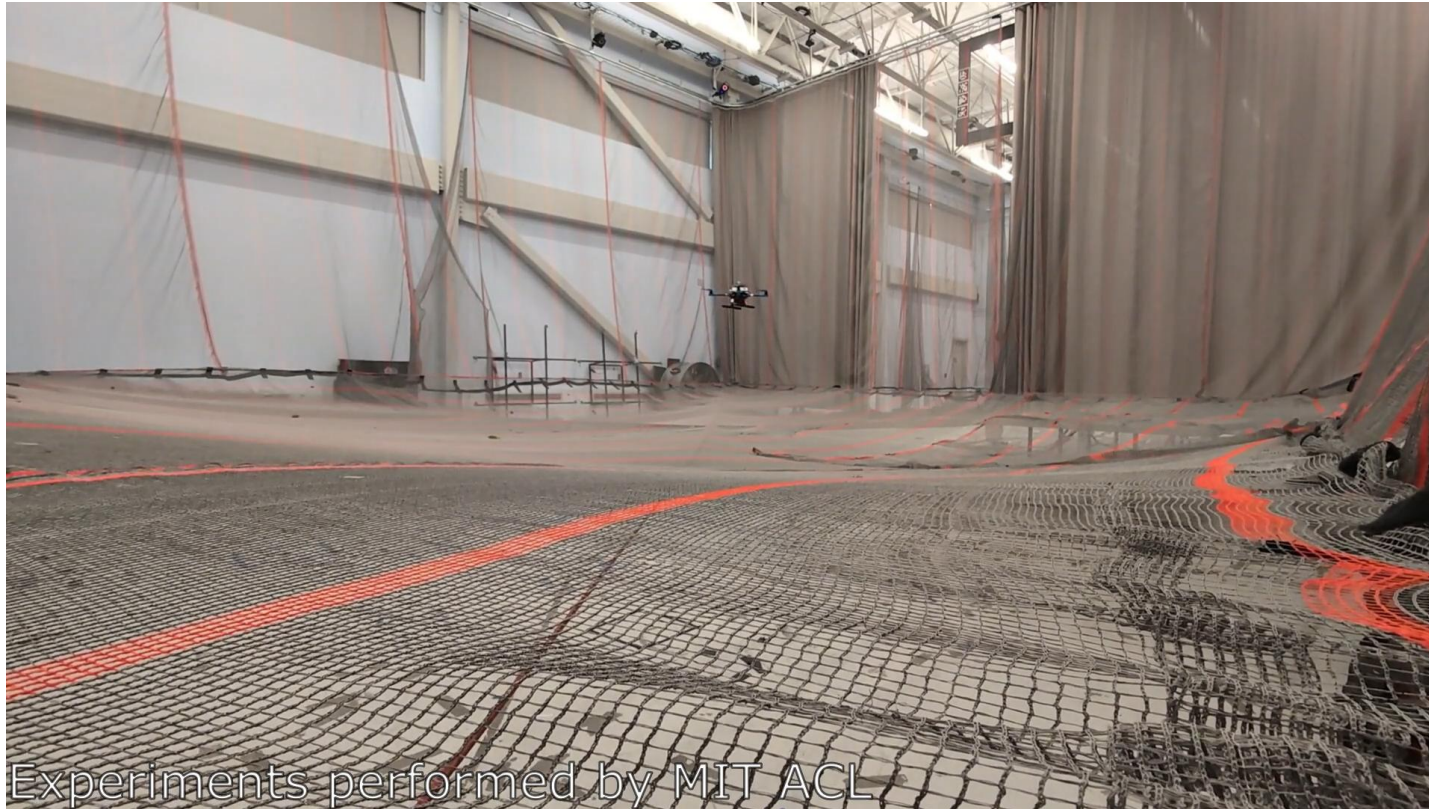


AVI

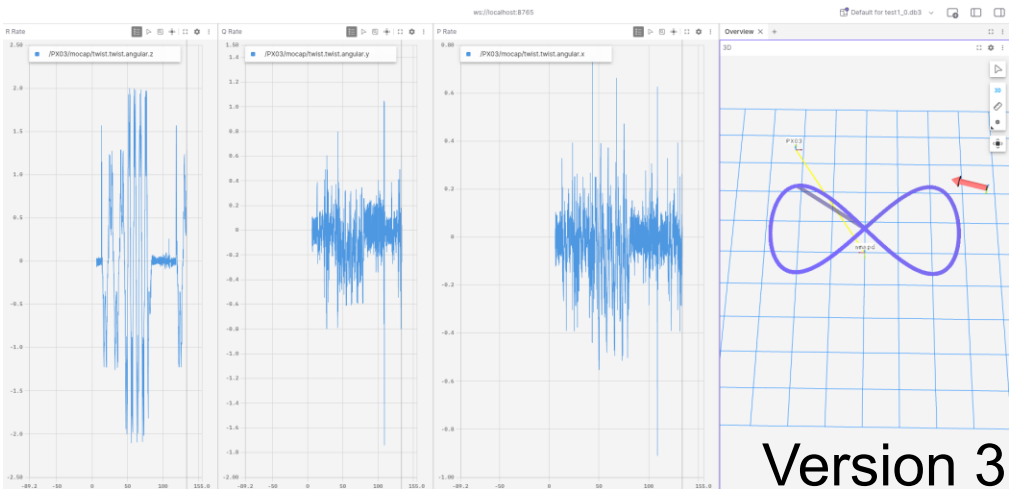
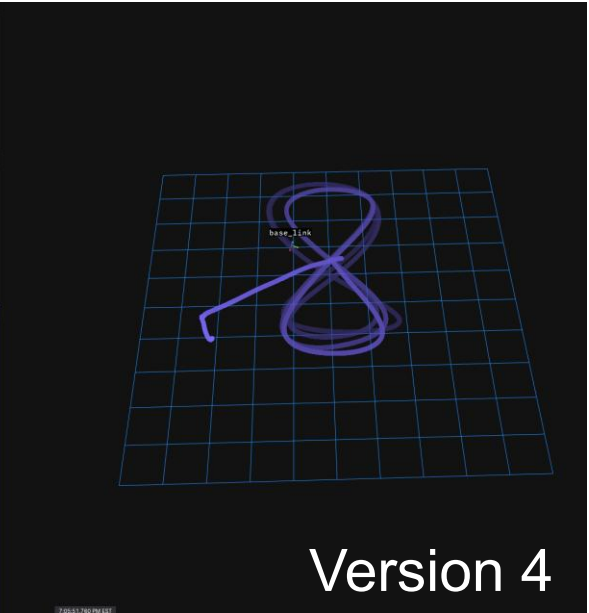
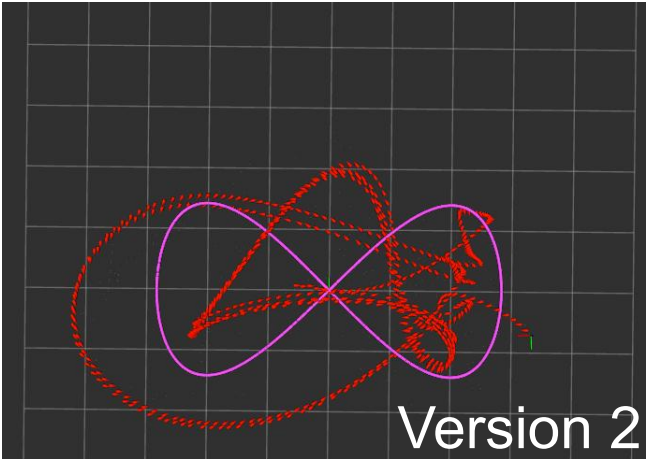
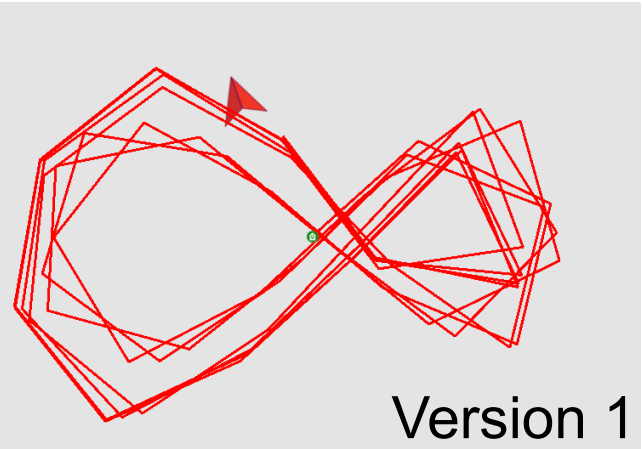
SCOPE OF MY WORK

- Debugging previous PX4 Implementation
- Building visualizations tools in Foxglove
- Designing a custom controller library in PX4
 - PID
 - RSLQR
 - OBLTR
 - AAOBLTR
- Obstacle Avoidance Analytical CBF
- HLOC Stack

DEBUGGING PREVIOUS PX4 IMPLEMENTATION

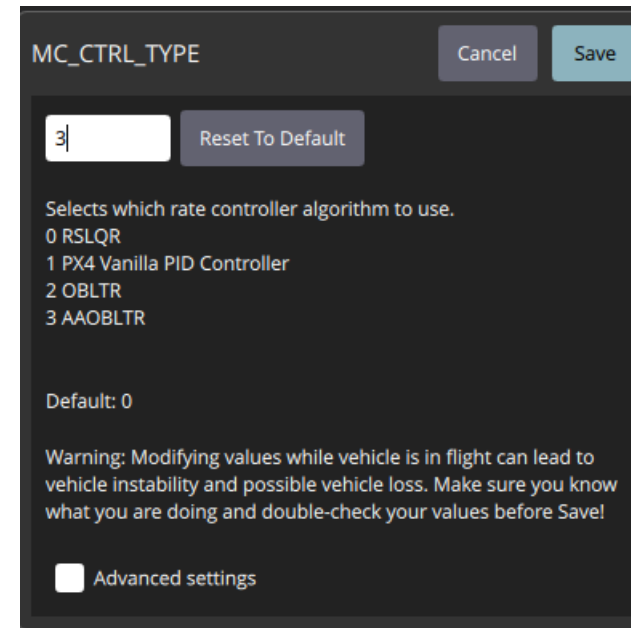
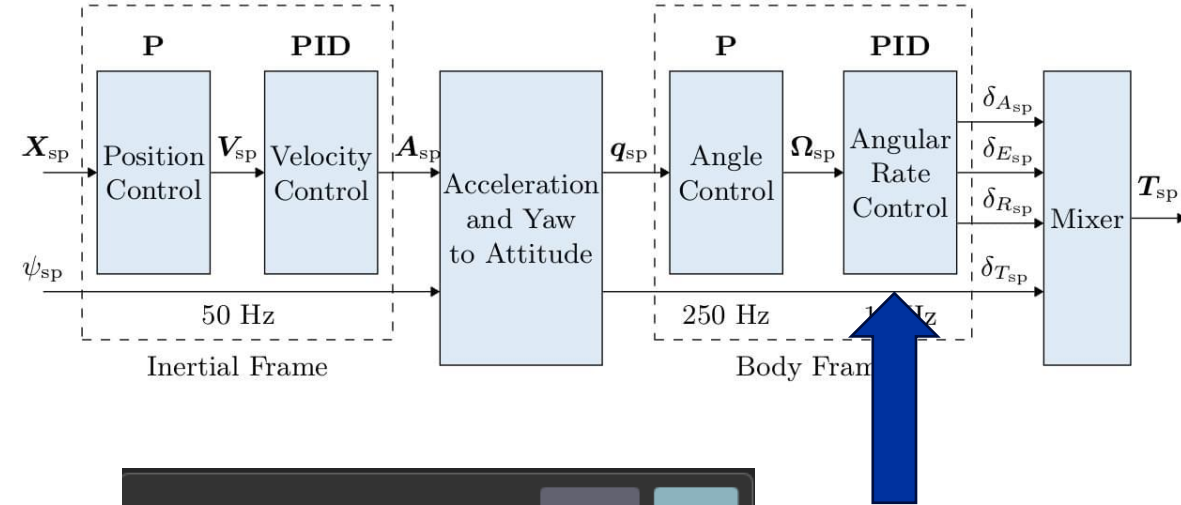


FOXGLOVE VISUALIZATION TOOLS



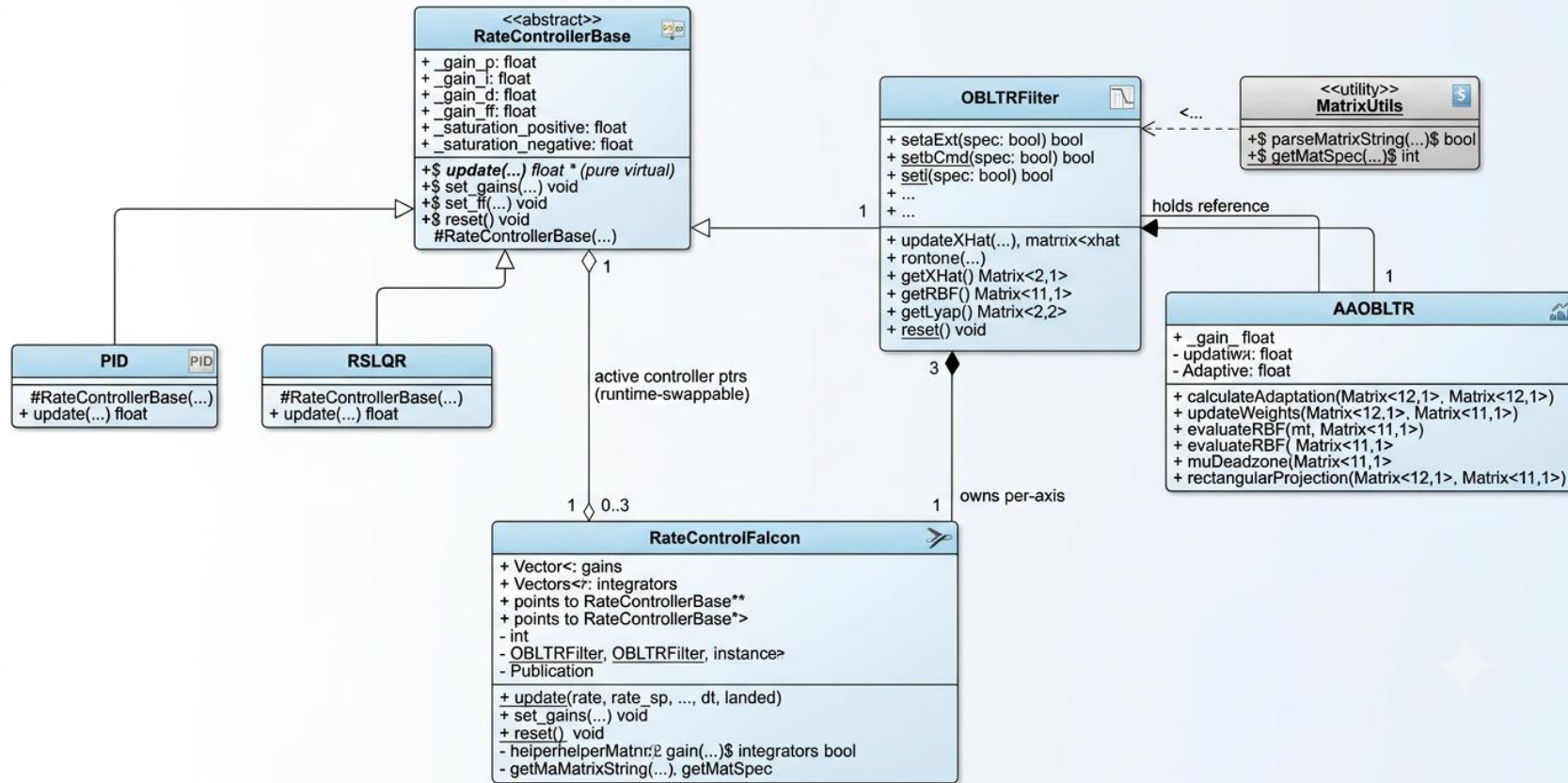
CUSTOM RATE CONTROLLER LIBRARY IN PX4

- For controller implementation
 - Working in the Angular Rate Control
- Forking and building custom PX4 Firmware
- Added new message types
 - Debugging
- Customized QGC to add custom fields



CUSTOM RATE CONTROLLER LIBRARY IN PX4

FALCON RATE CONTROL CLASS DIAGRAM



ADAPTIVE AUGMENTED OBSERVER-BASED CONTROL WITH LOOP TRANSFER RECOVERY (AAOBLTR)

- OBLTR:
 - A control method that estimates information that the drone can't directly senses, then uses it to fly as smoothly and reliably as if it could measure everything perfectly
- AAOBLTR:
 - An upgraded version that also adjusts itself on the fly, so the drone keeps flying well if conditions change

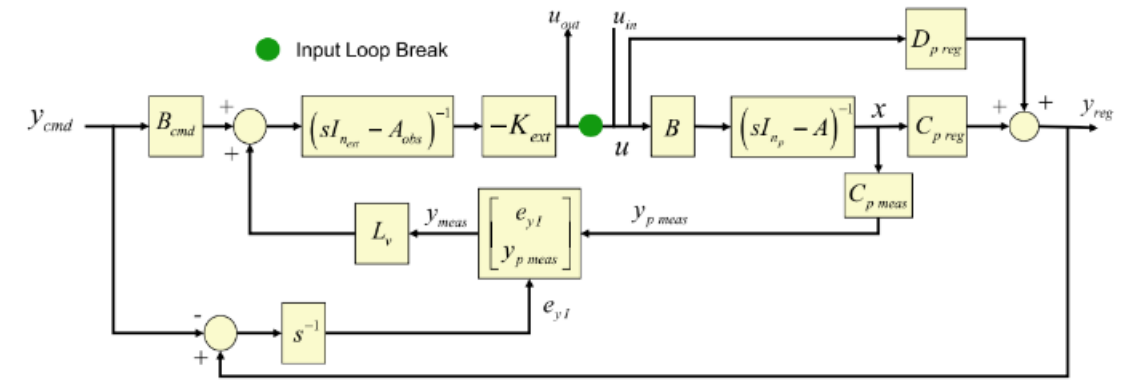
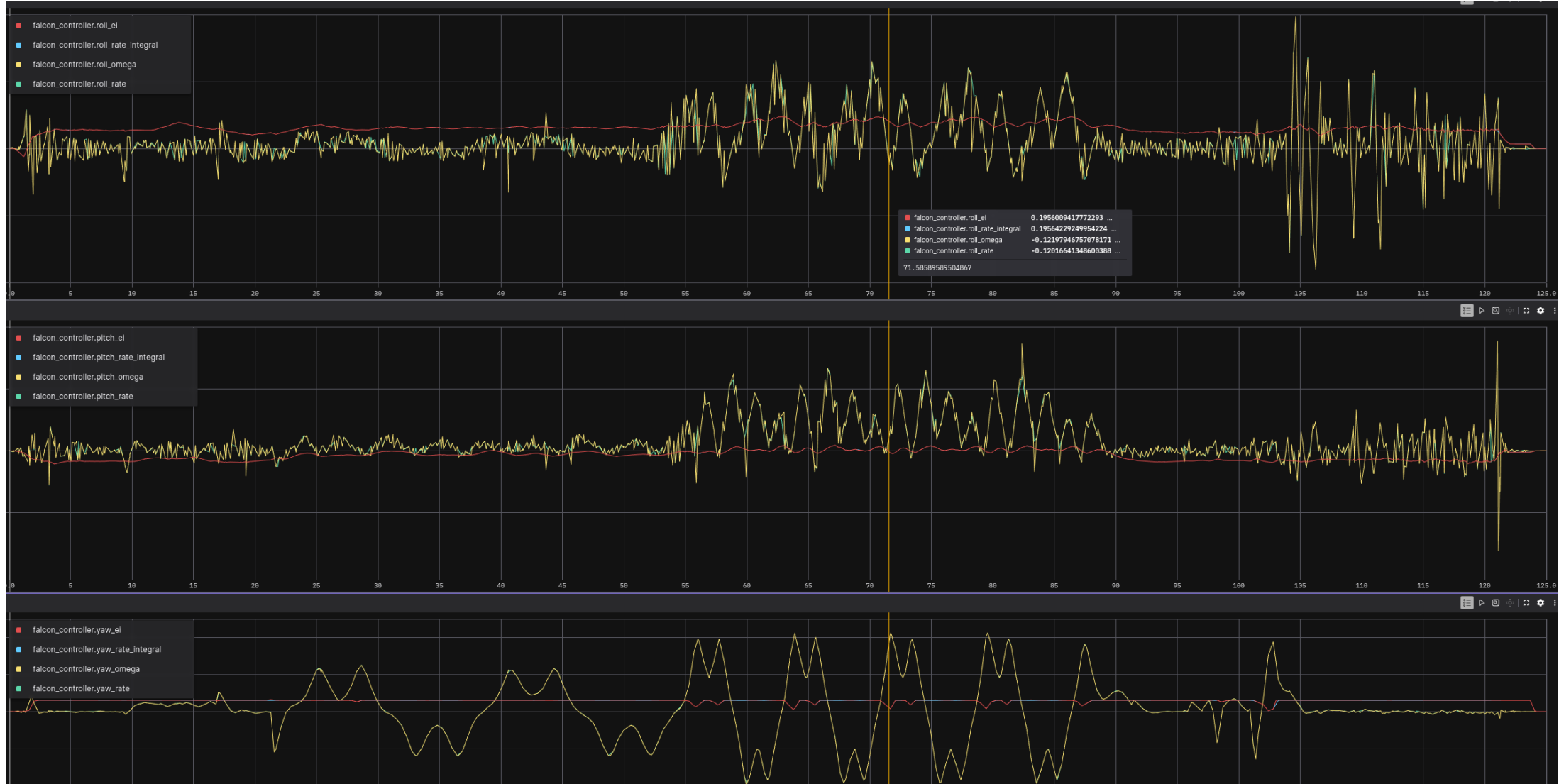


Fig. 6.13 Input loop gain calculation block diagram

TUNING OBLTR / AAOBLTR



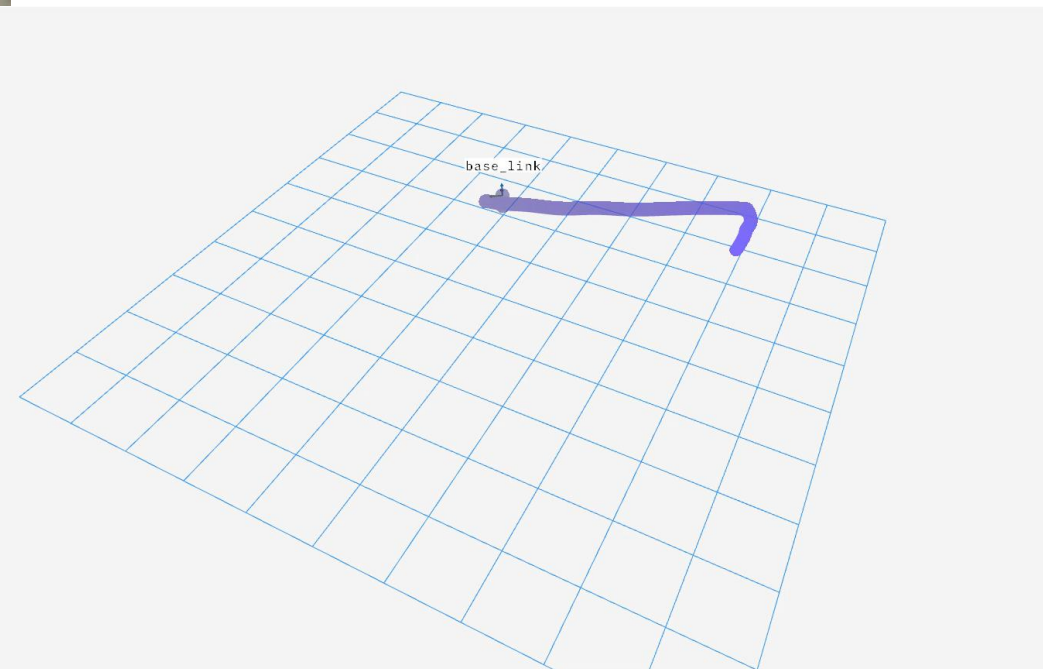
- $E_i == \text{Rate Integral}$
- $\Omega == \text{Pitch Rate}$

TUNING OBLTR / AAOBLTR



- Rates are tracking rate setpoints well

AAOBLTR DOING A FIGURE 8



OBSTACLE AVOIDANCE ANALYTICAL CBF SIMULATION

- Designed an Analytical CBF to avoid an obstacle set in the path of the Figure 8 mission

$$\Delta \mathbf{p} = \mathbf{p} - \mathbf{o}$$

$$h = \|\Delta \mathbf{p}\|^2 - r^2$$

$$\psi_1 = 2 \Delta \mathbf{p} \cdot \mathbf{v} + \alpha_1 h$$

$$\mathbf{c} = 2 \Delta \mathbf{p}$$

$$d = 2 \|\mathbf{v}\|^2 + 2 \alpha_1 (\Delta \mathbf{p} \cdot \mathbf{v}) + \alpha_2 \psi_1$$

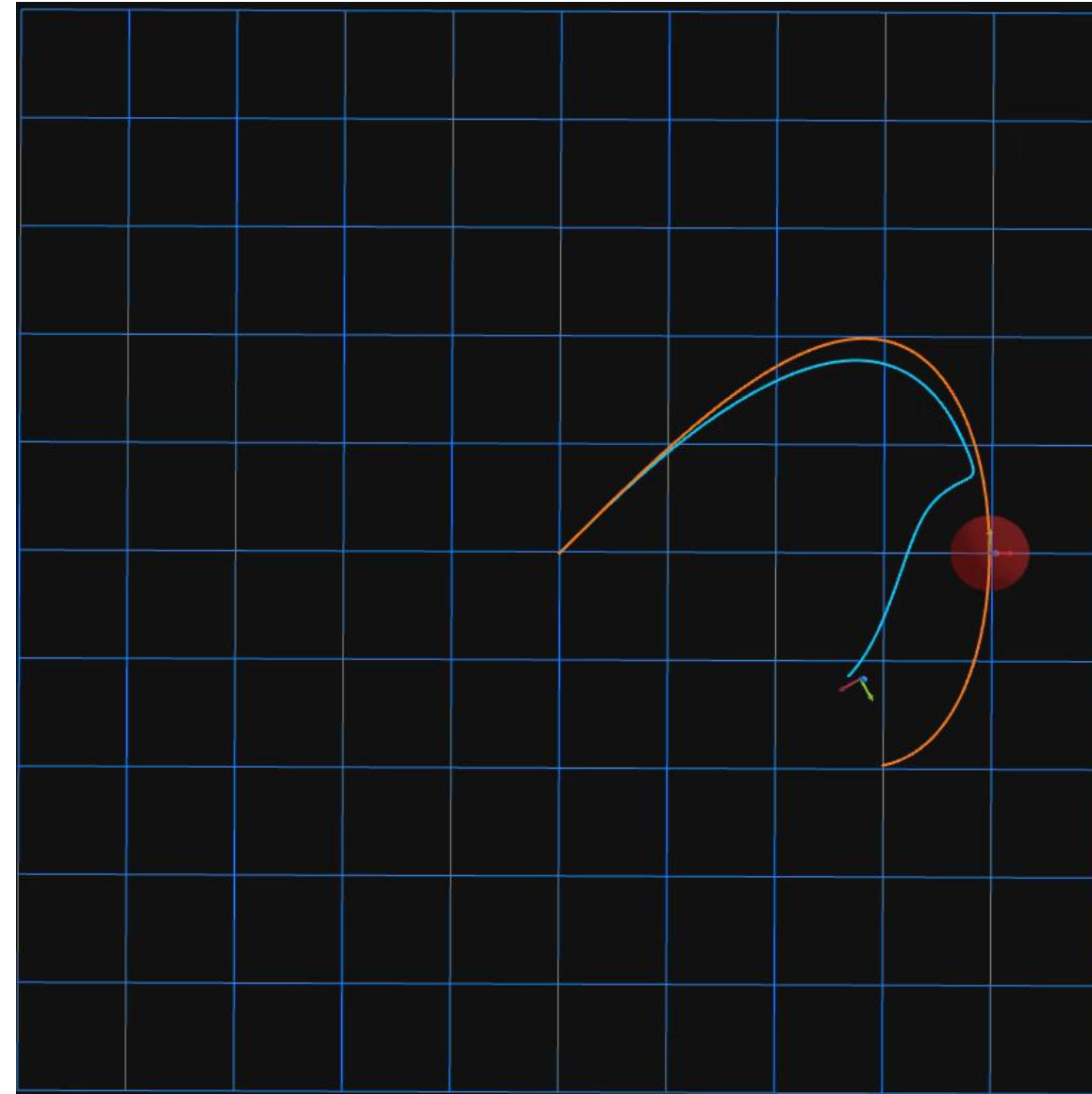
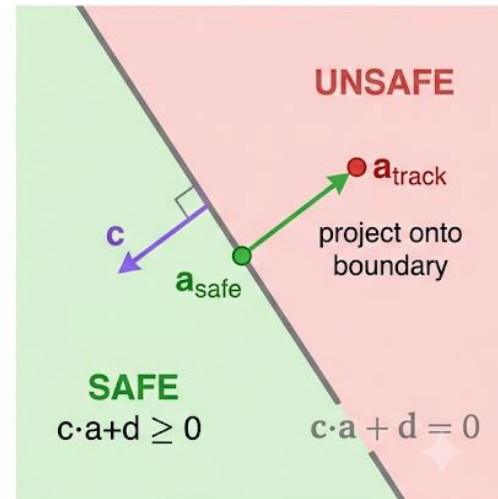
$$\text{margin} = \mathbf{c} \cdot \mathbf{a}_{\text{track}} + d$$

Solution:

$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}}, \text{margin} \geq 0$$

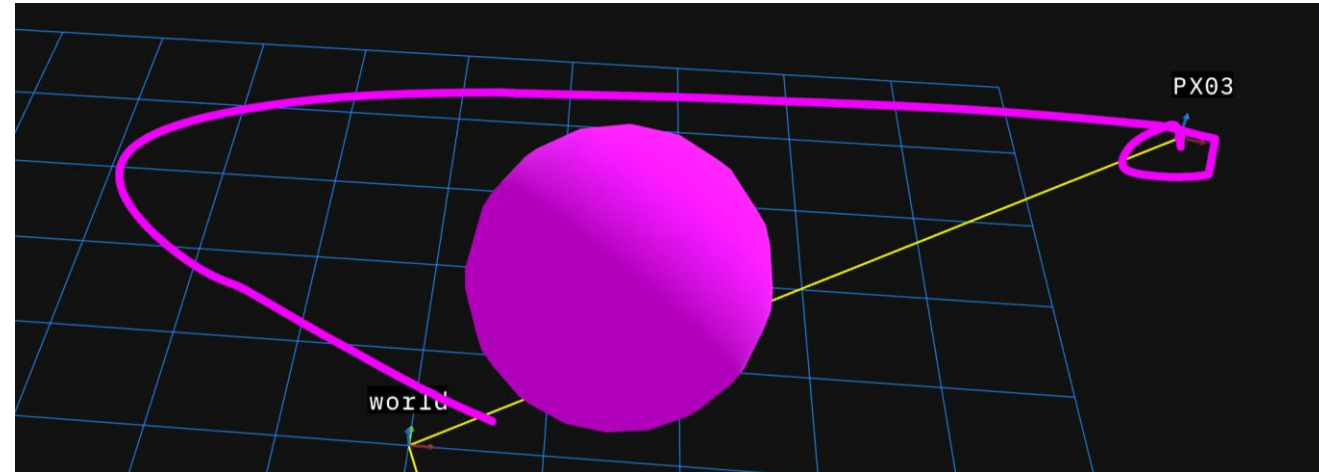
$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}} - \mathbf{c}(\text{margin}/\|\mathbf{c}\|^2), \text{margin} < 0$$

One active obstacle $\Rightarrow$ one linear inequality in $\mathbf{a}$
 $\Rightarrow$ closed form, no QP.



OBSTACLE AVOIDANCE ANALYTICAL CBF HARDWARE

- Had issues with ROS2 namespaces
 - Goals and the CBF augmentation were not being received
- Algorithm could not be tested on the hardware
- Good learning experience
 - Need a better pipeline for HITL testing



HLOC STACK

- Helped improve the boat HLOC perception stack
 - Increased inlier count significantly
 - Improved the performance of HLOC pose estimate
 - Used an off the shelf YOLOV8-N Detector Model



LESSONS LEARNED

Technical

- Hardware testing is hard
- Steep ramp-up on advanced controllers
- Visualization tools are essential

Professional

- First time with hard deadlines
- Fast-paced, experienced team

People & Culture

- In-office connections matter
- Great work culture boost productivity

